

AI FarmTwin – Predict, Simulate & Protect Smart Farms

1. Problem Statement / Problem Being Addressed

Modern farming depends heavily on timely decisions about crop health, irrigation, environmental conditions, and farm safety. However, farmers often lack a unified digital system that can provide a clear real-time view of their farm and help them understand how changing conditions may affect crops. Sensor data, crop observations, environmental information, and farm events are usually handled separately, making it difficult to identify risks early and take preventive action.

The problem addressed by **AI Farm Twin** is the lack of an accessible, centralized digital representation of a farm that can monitor crop conditions, organize plant-level information, identify potential risks, and support data-driven decision making. Large farms can contain multiple zones, sections, and thousands of plants, making manual monitoring time-consuming and difficult to manage.

AI FarmTwin addresses this challenge by creating a **Digital Twin of the farm** where the farm is divided into configurable zones and sections, and individual plants are digitally represented and organized into groups of ten. Environmental values such as temperature, humidity, soil moisture, and weather conditions can be entered and analyzed to determine crop health and risk levels. The platform also provides event detection, plant health monitoring, historical records, reports, and what-if simulation capabilities.

The current prototype demonstrates this concept through a software-only implementation using manually entered sensor values. This allows the complete workflow to be demonstrated without physical hardware while keeping the architecture ready for future IoT integration.

2. Proposed Solution

AI FarmTwin is a web-based smart agriculture platform that creates a digital representation of a physical farm and provides a centralized interface for monitoring, analysis, simulation, and decision support.

After registration and email verification, each user receives a **private farm environment** connected to their own **Firestore account**. The farmer can configure the farm, create multiple zones and sections, define crop information, and maintain plant records. The system automatically organizes every ten consecutive plants into a plant group with a shared physical address, making large farms easier to manage.

The main **Digital Twin** provides a top-down representation of each farm zone. Four sections—North, South, East, and West—are represented within the same continuous field, with plant positions corresponding to actual plant records. Manual sensor values such as temperature, humidity, and soil moisture can be entered and processed by the system to determine crop health, stress, and risk conditions.

The platform also includes Plant Management, Crop Health Analysis, Event Detection, Intrusion History, Reports, and a What-If Simulator. The simulator allows users to test hypothetical environmental changes and observe how the farm's health indicators could respond without permanently changing the actual farm data.

For the current hackathon demonstration, sensor values are manually entered to demonstrate the complete software workflow. In a real deployment, these inputs can be replaced or supplemented by **ESP32/IoT-connected physical sensors**, allowing live farm conditions to automatically update the Digital Twin.

3. Key Features

- 1) Interactive Farm Digital Twin
- 2) Multiple configurable farm zones
- 3) Four sections per zone: North, South, East and West
- 4) Real plant records represented on the digital field
- 5) Plant grouping — 10 plants per group with shared address
- 6) Crop health and risk monitoring
- 7) Temperature, humidity and soil-moisture monitoring
- 8) Manual sensor-data input for prototype demonstration
- 9) Event Detection for animal, human and vehicle activity
- 10) Event location and plant-group identification
- 11) Intrusion/event history
- 12) Plant health filtering and search
- 13) Crop health comparison
- 14) What-If Simulation
- 15) Temporary simulation without modifying real farm data
- 16) Automated reports and PDF export
- 17) Persistent user-specific farm data
- 18) Firebase authentication and email verification
- 19) Password reset
- 20) Private data for every registered user
- 21) Responsive web interface
- 22) Architecture prepared for future IoT sensor integration

4. Technology Stack

- Frontend
- React.js

- JavaScript
- HTML5
- CSS3
- Responsive web design

Backend / Cloud

- Firebase Firestore
- UID-based user data architecture
- Cloud-based persistent storage

Data & Visualization

- Interactive Digital Twin visualization
- Dynamic plant and zone rendering
- Health/risk calculations
- Charts and analytical dashboards
- PDF report generation using jsPDF

Future Hardware Integration

- ESP32 / NodeMCU
- Temperature and humidity sensors
- Soil-moisture sensors
- Other agricultural/environmental sensors
- IoT-to-cloud communication

5. AI Technologies

For the current prototype, the intelligence is primarily implemented through **data-driven health/risk analysis, automated decision logic, event analysis, and What-If simulation** rather than depending on a physical AI device.

AI/Intelligent Components

- **Crop Health Analysis** – evaluates environmental conditions and determines health/risk status.
- **Risk Detection** – identifies warning, high-risk and critical conditions based on farm parameters.
- **What-If Simulation** – evaluates how hypothetical changes in environmental conditions may affect farm health.

- **Event Analysis** – determines the affected zone, section and nearest plant group from event coordinates.
 - **Historical Analysis** – allows changes in plant/farm conditions to be tracked over time.
 - **Decision Support** – converts farm data into understandable health and risk information for the farmer.
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6. APIs / Services Used

Current prototype:

- **Firestore** – secure, persistent storage of each user's farm data.
- **jsPDF** – generation of downloadable farm reports.

AI/API clarification:

The current prototype does **not depend on a paid external AI API for its core farm monitoring functionality**. The intelligence demonstrated in the prototype is implemented within the application using farm-data analysis, health/risk logic, event processing and simulation.

The architecture is designed so that future AI/ML APIs or custom models can be integrated without rebuilding the entire platform.

7. Hardware Conversion / Future Scope

AI FarmTwin is currently demonstrated as a software prototype, but it is designed to be converted into a real IoT-based agricultural monitoring system.

In the current demonstration:

Manual Sensor Input → FarmTwin → Analysis → Digital Twin → Alerts/Reports

In the real-world version:

Physical Sensors → ESP32/IoT Gateway → Firestore/Cloud → AI FarmTwin → Digital Twin → Analysis → Alerts/Reports

Sensors deployed throughout different farm zones could continuously provide temperature, humidity, soil moisture and other environmental measurements. The ESP32 could transmit these readings to the cloud, where AI FarmTwin would automatically update the corresponding zone and section of the Digital Twin.

This makes the current website a software foundation for a future smart-farming IoT system, rather than a prototype that is limited to manual data entry.

Our current prototype demonstrates the complete digital-twin workflow using manually entered sensor values. The same input layer can later be replaced by ESP32-connected physical sensors, enabling the Digital Twin to operate with live real-world farm data